Lecture Scribe_16

L15_L17.pdf
PDF

Generate a formal, academic lecture scribe suitable for exam preparation using ONLY the provided context (lecture slides / transcript / textbook excerpt). While writing the scribe, reason step by step through definitions, assumptions, theorems, proofs, derivations, and example solutions exactly as presented in the context, and include that step-by-step reasoning explicitly in the final scribe so the logical flow is clear for revision; explicitly state all definitions and assumptions, and for every proof, derivation, or worked example in the context, reproduce it in a numbered step-by-step exam style with no missing steps. Do not introduce any new material, alternative reasoning paths, extra examples, or external facts beyond what is explicitly contained in the provided context. Maintain a formal university-level tone optimized for exam revision, use clear headings and subheadings, and ensure every statement is traceable to the provided context.

Lecture Scribe (Exam Revision Format)

CSE400 – Fundamentals of Probability in Computing

Lecture L15_16_17_18: Joint Random Variables + Joint Distributions + Examples

Source used: Lecture PDF only

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1. Introduction: Why Study Joint Random Variables?

In science and in real life, we are often interested in **two (or more) random variables at the same time**. The lecture motivates this using several examples:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate

- Number of Facebook friends and age of members

The lecture also provides application-driven examples:

Example 1: Smart Transportation

- $X =$ Vehicle Speed
- $Y =$ Traffic Density

Question posed in lecture:
Can speed be high when the traffic density is also high?
Answer in lecture: Yes / No (posed as a conceptual relationship question)

This motivates that X and Y may be **correlated random variables**.
Applications mentioned:

- Adaptive traffic signals
- Congestion prediction
- Autonomous vehicle routing

Example 2: Wireless Network Performance

- $X =$ Signal-to-Noise Ratio (SNR)
- $Y =$ Data Throughput

Reason given in lecture:
Throughput depends on SNR, but both vary due to fading, interference, and congestion.

Question posed in lecture:
If SNR is high, will throughput always be high?
Answer in lecture: Yes / No

Reasoning explicitly stated in lecture:
Network congestion (conditional relationships).

Applications mentioned:

- 5G scheduling
- Adaptive modulation
- Network planning

Example 3: Weather Prediction

- $X =$ Rainfall

- $Y =$ Temperature
  Application: weather forecasting.

Example 4: Drone Delivery

- $X =$ Wind Speed
- $Y =$ Delivery Time
  Application: logistics and UAV systems.

Example 5: Rolling Two Dice

- The lecture specifically mentions events such as:
  - $X + Y = 7$
  - $X > Y$

Conclusion stated in lecture:

This requires a **joint PMF**.

Key Conceptual Conclusion

The lecture states:

- In such situations, the random variables have a **joint distribution**.
- A joint distribution allows us to:
  1. Compute probabilities of events involving both variables.
  2. Understand the relationship between the variables.
- This is simplest when the variables are **independent**.
- When they are not independent, we use **covariance** and **correlation** as measures of the nature of dependence between them.

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2. Discrete Case – Joint PMF

2.1 Setup

Suppose X and Y are discrete random variables.

The lecture gives:

- $X \in \{x_1, x_2, \ldots, x_n\}$
- $Y \in \{y_1, y_2, \ldots, y_m\}$

Then the ordered pair (X, Y) takes values in the **product set**:

$$\{(x_1, y_1), (x_1, y_2), \ldots, (x_n, y_m)\}$$

This means every possible joint outcome is a pair formed by one value of X and one value of Y.

2.2 Definition of Joint PMF

Definition (Joint PMF)

The lecture defines the joint PMF as:

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

Meaning

This gives the probability that the **joint outcome occurs simultaneously**, i.e., both:

- $X = x_i$, and
- $Y = y_j$.

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3. Joint PMF Table Representation

The lecture states that the joint PMF is organized in a **table**, where:

- rows correspond to values of one variable,
- columns correspond to values of the other variable,
- each cell contains the joint probability $p(x_i, y_j)$.

3.1 Key Properties of the Joint PMF

The lecture explicitly lists the following properties:

Property 1: Bounds

$$0 \leq p(x, y) \leq 1$$

Property 2: Total Probability

$$\sum_i \sum_j p(x_i, y_j) = 1$$

This means the probabilities across **all possible joint outcomes** must sum to 1.

4. Example: Rolling Two Dice (Discrete Joint PMF Example)

4.1 Experiment Setup

The lecture defines the experiment:

- Roll two fair dice.

Define:

- $X =$ value on the **first die**
- $T =$ **total (sum)** of both dice

4.2 Probability Structure

The lecture states:

- Each outcome (i, j) from the two dice has probability:

$$\frac{1}{36}$$

Therefore, the joint probability table of (X, T) can be constructed from the equally likely two-die outcomes.

4.3 Observations from the Lecture

The lecture explicitly states:

1. **Entries are either 0 or $\frac{1}{36}$**
2. **X and T are not independent**

4.4 Exam-Style Interpretation of the Table (Exactly as implied by lecture)

Step 1: Identify the variables

- X is the first die value.
- T is the sum of the two dice.

Step 2: Determine the sample space

Each ordered pair (i, j), where:

- $i \in \{1, 2, 3, 4, 5, 6\}$
- $j \in \{1, 2, 3, 4, 5, 6\}$

has probability $\frac{1}{36}$.

Step 3: Translate to (X, T)

For a fixed first die value $X = i$, the total is:

$$T = i + j$$

where j is the second die value.

Step 4: Determine valid joint outcomes

A joint pair $(X, T) = (i, t)$ is possible only if:

$$t = i + j \quad \text{for some } j \in \{1, 2, 3, 4, 5, 6\}$$

Equivalently,

$$j = t - i \in \{1, 2, 3, 4, 5, 6\}$$

Step 5: Assign probability

- If $t - i \in \{1, 2, 3, 4, 5, 6\}$, then exactly one second-die value produces that pair, so:

$$P(X = i, T = t) = \frac{1}{36}$$

- Otherwise:

$$P(X = i, T = t) = 0$$

Step 6: Conclusion

Hence the lecture's observation follows:

- all table entries are either 0 or $\frac{1}{36}$,
- and X and T are not independent.

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5. Continuous Case – Joint PDF

5.1 Analogy with the Discrete Case

The lecture explicitly states the following analogies:

- **Discrete values $\rightarrow$ Continuous intervals**

- **Joint PMF $\to$ Joint PDF**
- **Sums $\to$ Double integrals**

5.2 Setup

If:

- $X \in [a, b]$
- $Y \in [c, d]$

then the pair (X, Y) takes values in the **product region**:

$$[a, b] \times [c, d]$$

This is the rectangular region formed by the ranges of X and Y.

5.3 Definition of Joint PDF

Definition (Joint PDF)

A function $f(x, y)$ is called the **joint probability density function** of X and Y if:

$$P((X, Y) \text{ lies in a small rectangle of area } dx\, dy) = f(x, y)\, dx\, dy$$

This is the lecture's formal density interpretation.

5.4 Key Properties of the Joint PDF

The lecture explicitly lists:

Property 1: Nonnegativity

$$f(x, y) \geq 0$$

Property 2: Normalization

$$\iint f(x, y)\, dx\, dy = 1$$

The lecture also emphasizes:

- The joint PDF $f(x, y)$ **may take values greater than 1**.
- It is a **probability density**, **not a probability**.

6. Worked Example (Continuous Joint PDF)

6.1 Given Data

The lecture gives:

$$f(x, y) = 4xy, \qquad 0 < x < 1,\ 0 < y < 1$$

The lecture states that jointly, (X, Y) takes values in the **unit square**:

$$[0, 1]^2$$

Tasks in the lecture

1. Show that $f(x, y)$ is a valid joint PDF.
2. Visualize the event

$$A = \{X < 0.5,\ Y > 0.5\}$$

3. Compute $P(A)$.

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7. Example Solution – Step-by-Step (Exactly Following Lecture)

7.1 Step 1: Check Validity (Normalization)

The lecture computes:

$$\int_0^1 \int_0^1 4xy\, dx\, dy$$

Step-by-step derivation exactly in exam style

Step 1.1: Write the normalization integral

$$\int_0^1 \int_0^1 4xy\, dx\, dy$$

Step 1.2: Integrate with respect to x first

$$= \int_0^1 \left(\int_0^1 4xy\, dx \right) dy$$

Since y is constant with respect to x:

$$\int_0^1 4xy\,dx = 4y\int_0^1 x\,dx$$

$$= 4y\left[\frac{x^2}{2}\right]_0^1$$

$$= 4y\left(\frac{1}{2}\right) = 2y$$

So:

$$\int_0^1\int_0^1 4xy\,dx\,dy = \int_0^1 2y\,dy$$

Step 1.3: Integrate with respect to y

$$\int_0^1 2y\,dy = [y^2]_0^1$$

$$= 1^2 - 0^2 = 1$$

Conclusion from the lecture

Since:

- $f(x,y) > 0$, and
- total probability is 1,

therefore:

$$f(x,y) = 4xy$$

is a **valid joint PDF**.

7.2 Step 2: Compute $P(A)$

The event is:

$$A = \{X < 0.5,\ Y > 0.5\}$$

So the lecture computes probability over:

- $0 < x < 0.5$
- $0.5 < y < 1$

Step-by-step derivation exactly as presented

Step 2.1: Write the probability integral

$$P(A) = \int_0^{0.5} \int_{0.5}^{1} 4xy \, dy \, dx$$

Step 2.2: Integrate with respect to y first

$$= \int_0^{0.5} \left[2xy^2\right]_{0.5}^{1} dx$$

Step 2.3: Substitute the bounds

$$= \int_0^{0.5} 2x(1 - 0.25) \, dx$$

$$= \int_0^{0.5} 2x(0.75) \, dx$$

$$= \int_0^{0.5} \frac{3}{2} x \, dx$$

Step 2.4: Integrate with respect to x

$$= \left[\frac{3}{4}x^2\right]_0^{0.5}$$

$$= \frac{3}{4}(0.5)^2$$

$$= \frac{3}{4} \cdot \frac{1}{4}$$

$$= \frac{3}{16}$$

Final Answer

$$P(A) = \frac{3}{16}$$

This is the exact result shown in the lecture.

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8. Joint Cumulative Distribution Function (Joint CDF)

8.1 Notation

Suppose X and Y are jointly distributed random variables.

The lecture states the notation:

$$X < x,\ Y < y$$

means the event:

$$\{X < x \text{ and } Y < y\}$$

This is the event that **both inequalities occur simultaneously**.

8.2 Definition of Joint CDF

Definition

The joint cumulative distribution function is defined as:

$$F(x, y) = P(X < x,\ Y < y)$$

This is the probability that:

- X is less than x, and
- Y is less than y.

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9. Joint CDF – Continuous Case

If X and Y are continuous random variables with joint density $f(x, y)$ over $[a, b] \times [c, d]$, then the lecture states that the joint CDF is obtained by **integrating the density**:

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

This expresses cumulative probability over the rectangle from the lower-left corner of the support up to (x, y).

9.1 Recovering the Joint PDF from the Joint CDF

The lecture explicitly states:

Since there are two variables, we use **partial derivatives**:

$$f(x, y) = \frac{\partial^2}{\partial x\, \partial y} F(x, y)$$

This is the recovery formula from the continuous joint CDF.

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10. Joint CDF – Discrete Case

If X and Y are discrete random variables with joint PMF $p(x_i, y_j)$, then the lecture states that the joint CDF is obtained by **summing probabilities**:

$$F(x, y) = \sum_{x_i < x} \sum_{y_j < y} p(x_i, y_j)$$

This means we add all joint PMF entries whose:

- row values satisfy $x_i < x$, and
- column values satisfy $y_j < y$.

10.1 Interpretation (Exactly as Given in Lecture)

The lecture gives the following interpretation:

- Add all probabilities in the table
- For rows with $x_i < x$
- And columns with $y_j < y$

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11. Properties of the Joint CDF

The lecture states that the joint CDF $F(x, y)$ of X and Y must satisfy several properties.

11.1 Property 1: Non-decreasing

$$F(x, y) \text{ is non-decreasing}$$

Meaning stated in lecture

If x or y increase, then $F(x, y)$ must:

- stay constant, or
- increase.

11.2 Property 2: Lower-left Limit

The lecture states:

$$F(x, y) = 0 \text{ at the lower-left of the joint range}$$

If the lower-left corner is $(-\infty, -\infty)$, then:

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

This expresses that cumulative probability before the support begins is zero.

11.3 Property 3: Upper-right Limit

The lecture states:

$$F(x, y) = 1 \text{ at the upper-right of the joint range}$$

If the upper-right corner is (∞, ∞), then:

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

This expresses that cumulative probability over the entire joint range equals 1.

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12. Consolidated Exam Revision Summary

12.1 Core Definitions to Memorize

Discrete Joint PMF

$$p(x_i, y_j) = P(X = x_i,\ Y = y_j)$$

Continuous Joint PDF

$$P((X, Y) \text{ lies in a small rectangle of area } dx\, dy) = f(x, y)\, dx\, dy$$

Joint CDF

$$F(x, y) = P(X < x,\ Y < y)$$

12.2 Key Discrete Facts

- (X, Y) takes values in the **product set** of values of X and Y.
- Joint PMF table entries represent simultaneous outcomes.
- Properties:

$$0 \leq p(x, y) \leq 1$$

$$\sum_i \sum_j p(x_i, y_j) = 1$$

12.3 Key Continuous Facts

- (X, Y) takes values in a **product region** such as $[a, b] \times [c, d]$.
- Properties:

$$f(x, y) \geq 0$$

$$\iint f(x, y)\, dx\, dy = 1$$

- $f(x, y)$ may be greater than 1 because it is a **density**, not a probability.

12.4 Joint CDF Formulas

Continuous

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

Recover density

$$f(x, y) = \frac{\partial^2}{\partial x\, \partial y} F(x, y)$$

Discrete

$$F(x, y) = \sum_{x_i < x} \sum_{y_j < y} p(x_i, y_j)$$

12.5 Worked Example Final Results

For:

$$f(x,y) = 4xy, \quad 0 < x < 1,\ 0 < y < 1$$

- Validity check:

$$\int_0^1 \int_0^1 4xy\,dx\,dy = 1$$

  Hence valid joint PDF.
- For:

$$A = \{X < 0.5,\ Y > 0.5\}$$

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy\,dy\,dx = \frac{3}{16}$$

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13. Final Exam-Oriented Closing Note

This lecture establishes the foundational framework for analyzing **two random variables simultaneously**:

1. **Discrete case:** Joint PMF and table representation
2. **Continuous case:** Joint PDF and double integrals
3. **Cumulative form:** Joint CDF in both continuous and discrete settings
4. **Worked examples:**
   - Two-dice joint PMF structure
   - Continuous joint PDF example with full normalization and event probability computation

All formulas, interpretations, and derivations above are reproduced strictly from the provided lecture material and organized in step-by-step exam style for revision.

If you want, I can also make this into a "**night-before-exam ultra-condensed 2-page revision sheet**" in the *same strict no-extra-material style*.